

FULL NETWORK CONFIGURATION (CISCO CLI)

1. ROUTER CONFIGURATION

```
enable
configure terminal
hostname STADIUM-RTR

interface g0/0/1
description TO-ISP
ip address 200.10.10.2 255.255.255.252
no shutdown

interface g0/0/0
description TO-FIREWALL
ip address 172.16.0.1 255.255.255.252
no shutdown

ip route 0.0.0.0 0.0.0.0 200.10.10.1

ip route 10.0.0.0 255.255.0.0 172.16.0.2
ip route 192.168.100.0 255.255.252.0 172.16.0.2

end
write memory
```

2. ASA FIREWALL CONFIGURATION

```
enable
configure terminal
hostname STADIUM-ASA

interface g1/1
nameif outside
```

```
security-level 0
ip address 172.16.0.2 255.255.255.252
no shutdown
```

```
interface g1/2
nameif inside
security-level 100
ip address 10.0.0.1 255.255.255.0
no shutdown
```

```
route outside 0.0.0.0 0.0.0.0 172.16.0.1
```

```
object network INSIDE-NET
subnet 10.0.0.0 255.255.0.0
nat (inside,outside) dynamic interface
```

```
access-list BLOCK_PUBLIC extended deny ip 192.168.100.0 255.255.252.0 10.0.0.0
255.255.0.0
access-list BLOCK_PUBLIC extended permit ip any any
access-group BLOCK_PUBLIC in interface inside
```

```
write memory
```

3. CORE SWITCH (LAYER 3)

```
enable
configure terminal
hostname CORE-SW
```

```
ip routing
```

```
vlan 10
vlan 20
vlan 30
vlan 40
vlan 50
vlan 60
vlan 70
vlan 100
```

```
interface vlan 10
```

```
ip address 10.0.10.1 255.255.255.0
no shutdown
```

```
interface vlan 20
ip address 10.0.20.1 255.255.254.0
no shutdown
```

```
interface vlan 30
ip address 10.0.30.1 255.255.255.0
no shutdown
```

```
interface vlan 40
ip address 10.0.40.1 255.255.255.0
no shutdown
```

```
interface vlan 50
ip address 10.0.50.1 255.255.255.0
no shutdown
```

```
interface vlan 60
ip address 10.0.60.1 255.255.255.0
no shutdown
```

```
interface vlan 70
ip address 10.0.70.1 255.255.255.0
no shutdown
```

```
interface vlan 100
ip address 192.168.100.1 255.255.252.0
no shutdown
```

```
interface fa0/1
no switchport
ip address 10.0.0.2 255.255.255.0
no shutdown
```

```
ip route 0.0.0.0 0.0.0.0 10.0.0.1
```

```
interface range fa0/2 - 4
switchport mode trunk
no shutdown
```

```
end
write memory
```

4. DIST-SW1

```
enable
configure terminal
hostname DIST-SW1

vlan 10
vlan 20
vlan 70

interface fa0/1
switchport mode trunk
switchport trunk allowed vlan 10,20,70
no shutdown

interface fa0/2
switchport mode trunk
switchport trunk allowed vlan 10,20,70
no shutdown

end
write memory
```

5. DIST-SW2

```
enable
configure terminal
hostname DIST-SW2

vlan 30
vlan 40
vlan 50

interface fa0/1
switchport mode trunk
switchport trunk allowed vlan 30,40,50
no shutdown
```

```
interface fa0/2
switchport mode trunk
switchport trunk allowed vlan 30,40,50
no shutdown

end
write memory
```

6. DIST-SW3

```
enable
configure terminal
hostname DIST-SW3

vlan 100

interface fa0/1
switchport mode trunk
switchport trunk allowed vlan 100
no shutdown

interface fa0/2
switchport mode trunk
switchport trunk allowed vlan 100
no shutdown

end
write memory
```

7. ACCESS SWITCH 1

```
enable
configure terminal
hostname ACC-SW1

vlan 10
vlan 20
```

vlan 70

interface fa0/1
switchport mode trunk
no shutdown

interface range fa0/2 - 5
switchport mode access
switchport access vlan 10

interface range fa0/6 - 10
switchport mode access
switchport access vlan 20

interface range fa0/11 - 15
switchport mode access
switchport access vlan 70

interface range fa0/2 - 15
no shutdown

end
write memory

8. ACCESS SWITCH 2

enable
configure terminal
hostname ACC-SW2

vlan 30
vlan 40
vlan 50

interface fa0/1
switchport mode trunk
no shutdown

interface range fa0/2 - 6
switchport mode access
switchport access vlan 30

```
interface range fa0/7 - 10
switchport mode access
switchport access vlan 40
```

```
interface range fa0/11 - 15
switchport mode access
switchport access vlan 50
```

```
interface range fa0/2 - 15
no shutdown
```

```
end
write memory
```

9. ACCESS SWITCH 3

```
enable
configure terminal
hostname ACC-SW3
```

```
vlan 100
```

```
interface fa0/1
switchport mode trunk
no shutdown
```

```
interface range fa0/2 - 10
switchport mode access
switchport access vlan 100
no shutdown
```

```
end
write memory
```

10. DHCP CONFIG (CORE SWITCH)

```
configure terminal
```

service dhcp

ip dhcp excluded-address 10.0.10.1
ip dhcp excluded-address 10.0.20.1
ip dhcp excluded-address 10.0.30.1
ip dhcp excluded-address 10.0.40.1
ip dhcp excluded-address 10.0.50.1
ip dhcp excluded-address 10.0.60.1
ip dhcp excluded-address 10.0.70.1
ip dhcp excluded-address 192.168.100.1

ip dhcp pool VLAN10
network 10.0.10.0 255.255.255.0
default-router 10.0.10.1

ip dhcp pool VLAN20
network 10.0.20.0 255.255.254.0
default-router 10.0.20.1

ip dhcp pool VLAN30
network 10.0.30.0 255.255.255.0
default-router 10.0.30.1

ip dhcp pool VLAN40
network 10.0.40.0 255.255.255.0
default-router 10.0.40.1

ip dhcp pool VLAN50
network 10.0.50.0 255.255.255.0
default-router 10.0.50.1

ip dhcp pool VLAN60
network 10.0.60.0 255.255.255.0
default-router 10.0.60.1

ip dhcp pool VLAN70
network 10.0.70.0 255.255.255.0
default-router 10.0.70.1

ip dhcp pool PUBLIC
network 192.168.100.0 255.255.252.0
default-router 192.168.100.1

end

write memory

FINAL RESULT

- Full 3-tier architecture
- VLAN segmentation
- DHCP working
- Inter-VLAN routing via Core
- NAT + security via ASA
- Internet via Router